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GATE ECE · CONTROL SYSTEMS · CHAPTER-WISE PYQS

Time Response Analysis

GATE ECE Control Systems · Previous Year Questions with Answer Key

59	34	1988 – 2024	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	22 questions
Marks 2	32 questions
Marks 5	2 questions
Marks 8	2 questions
Marks 10	1 question
Answer key	at the end

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1 **Solve**
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2 **Check**
The answer key is on the last pages.

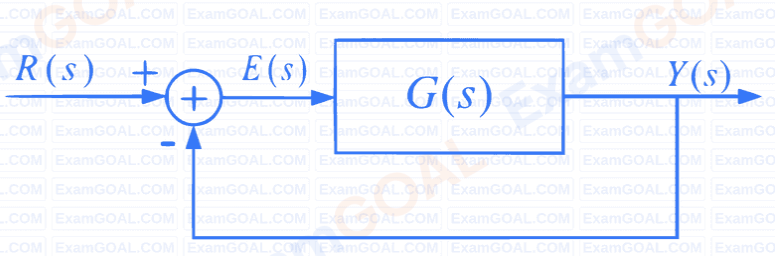
3 **Understand**
Scan the QR code beside any question for its full solution.

Marks 1 22 questions

Q1. GATE ECE 2024

Open solution

In the feedback control system shown in the figure below $G(s) = \dfrac{6}{s(s+1)(s+2)}$.



$R(s), Y(s)$, and $E(s)$ are the Laplace transforms of $r(t), y(t)$, and $e(t)$, respectively. If the input $r(t)$ is a unit step function, then _____

(A) $\lim_{t\to\infty} e(t) = 0$

(B) $\lim_{t\to\infty} e(t) = \dfrac{1}{3}$

(C) $\lim_{t\to\infty} e(t) = \dfrac{1}{4}$

(D) $\lim_{t\to\infty} e(t)$ does not exist, $e(t)$ is oscillatory

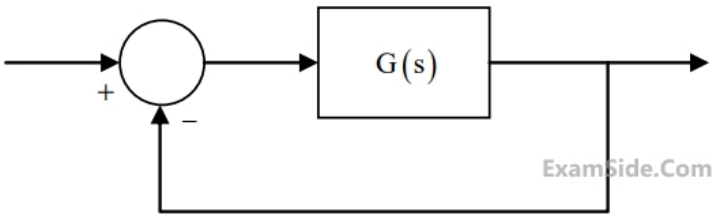
Q2. GATE ECE 2017 Set 1

Open solution

The open loop transfer function

$$G(s) = \frac{(s+1)}{s^p(s+2)(s+3)}$$

Where p is an integer, is connected in unity feedback configuration as shown in figure.



Given that the steady state error is zero for unit step input and is 6 for unit ramp input, the value of the parameter p is _____.

Q3. GATE ECE 2016 Set 2

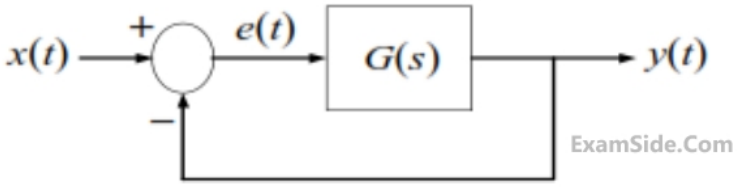
Open solution

The response of the system $G(s) = \frac{s-2}{(s+1)(s+3)}$ to the unit step input u(t) is y(t). The value of $\frac{dy}{dt}$ at $t = 0^+$ is _____.

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Q4. GATE ECE 2016 Set 3

For the unity feedback control system shown in the figure, the open-loop transfer function G(s) is given as $G(s) = \frac{2}{s(s+1)}$ .The steady state error e_{ss} due to a unit step input is



(A) 0 (B) 0.5 (C) 1.0 (D) ∞

Open solution

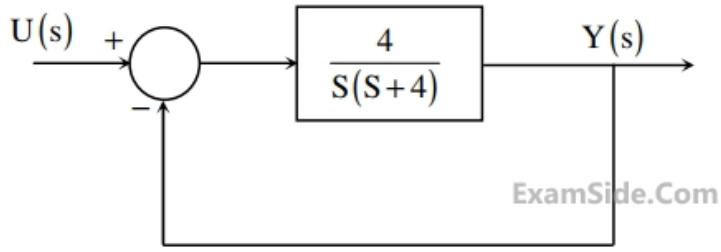
Q5. GATE ECE 2015 Set 2

A unity negative feedback system has an open–loop transfer function $G(s) = \frac{K}{s(s+1)}$.The gain K for the system to have a damping ratio of 0.25 is _____.

Open solution

Q6. GATE ECE 2014 Set 4

For the second order closed-loop system shown in the figure, the natural frequency (in rad/s) is



(A) 16 (B) 4 (C) 2 (D) 1

Open solution

Q7. GATE ECE 2014 Set 2

The natural frequency of an undamped second-order system is 40 rad/s. If the system is damped with a damping ratio 0.3, the damped natural frequency in rad/s is _____.

Open solution

Q8. GATE ECE 2014 Set 3

The input $-3e^{2t}\ u(t)$, where u(t) is the unit step function, is applied to a system with transfer function $\frac{s-2}{s+3}$. If the initial value of the output is -2, then the value of the output at steady state is _____.

Open solution

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Q9. GATE ECE 2011

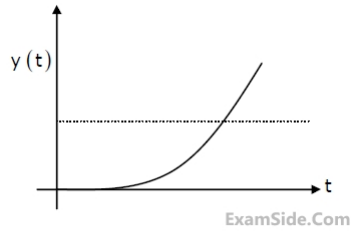
Open solution

The differential equation

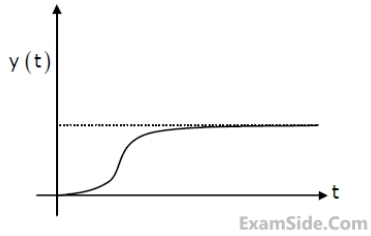
$$100\frac{d^2y}{dt^2} - 20\frac{dy}{dt} + y = x(t)$$

describes a system with an input x(t) and an output y(t). The system, which is initially relaxed, is excited by a unit step input. The output y(t) can be represented by the waveform

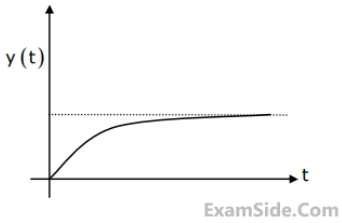
(A)



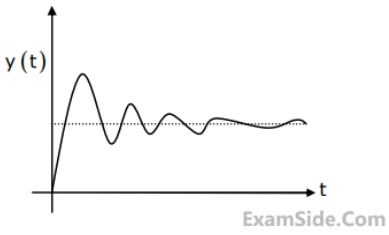
(B)



(C)



(D)



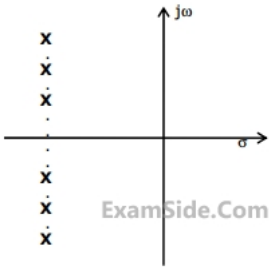
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Q10. GATE ECE 2008

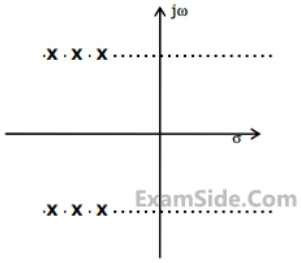
Step responses of a set of three second-order underdamped systems all have the same percentage overshoot. Which of the following diagrams represents the poles of the three systems?

Open solution

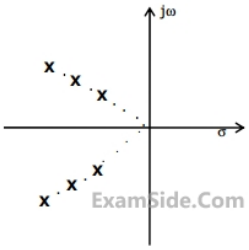
(A)



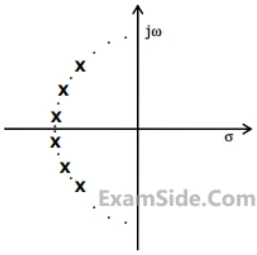
(B)



(C)



(D)



Q11. GATE ECE 2007

If the Laplace transform of a signal y(t) is $Y(s) = \frac{1}{s(s-1)}$, then its final value is:

(A) -1 (B) 0 (C) 1 (D) unbounded

Open solution

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Q12. GATE ECE 2002

Consider a system with the transfer function

$$G(s) = \frac{s+6}{Ks^2+s+6}$$

Its damping ratio will be 0.5 when the value of K is

(A) 2/6 (B) 3 (C) 1/6 (D) 6

Open solution

Q13. GATE ECE 2001

If the characteristic equation of a closed-loop system is $s^2 + 2s + 2 = 0$, then the system is

(A) overdamped (B) critically damped

(C) underdamped (D) undamped

Open solution

Q14. GATE ECE 1999

For a second-order system with the closed-loop transfer function

$$T(s) = \frac{9}{s^2+4s+9}$$

the settling time for 2% band in seconds is

(A) 1.5 (B) 2.0 (C) 3.0 (D) 4.0

Open solution

Q15. GATE ECE 1998

If $F(s) = \frac{\omega}{s^2+\omega^2}$, then the value of $\lim_{t\to\infty} f(t)$, $\{where\ F(s)\ is\ the\ L[f(t)]\}$

(A) cannot be determined (B) is zero

(C) is unity (D) is infinite

Open solution

Q16. GATE ECE 1998

Consider a unity feedback control system with open-loop transfer function $G(s) = \frac{K}{s(s+1)}$. The steady state error of the system due to a unit step input is

(A) zero (B) K (C) 1/K (D) infinite

Open solution

Q17. GATE ECE 1998

Consider a feedback control system with loop transfer function

$$G(s)H(s) = \frac{K(1+0.5s)}{s(1+s)(1+2s)}$$

The type of the closed loop system is

(A) 0 (B) 1 (C) 2 (D) 3

Open solution

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Q18. GATE ECE 1995

For a second order system, damping ratio (ξ) , is $0 < \xi < 1$,then the roots of the characteristic polynomial are

(A) Real but not equal
(B) Real and equal
(C) Complex conjugate
(D) Imaginary

Open solution

Q19. GATE ECE 1995

The final value theorem is used to find the

(A) steady state value of the system output
(B) initial value of the system output
(C) transient behaviour of the system output
(D) none of these

Open solution

Q20. GATE ECE 1995

If $L(f(t)) = \frac{2(s+1)}{s^2+2s+5}$ then f(0^+) and f(∞) are given by [Note: 'L' stands for 'Laplace Transform of']

(A) 0,2 respectively
(B) 2,0 respectively
(C) 0,1 respectively
(D) 2/5,0 respectively

Open solution

Q21. GATE ECE 1995

The step error coefficient of a system G(s)=$\frac{1}{(s+6)(s+1)}$ with unity feedback is

(A) 1/6
(B) ∞
(C) 0
(D) 1

Open solution

Q22. GATE ECE 1994

The poles of a continuous time oscillator are _____.

Open solution

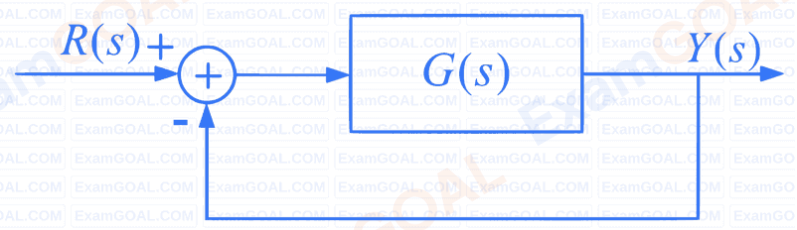
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Marks 2 32 questions

Q23. GATE ECE 2024

Open solution

Consider a unity negative feedback control system with forward path gain $G(s) = \frac{K}{(s+1)(s+2)(s+3)}$ as shown.



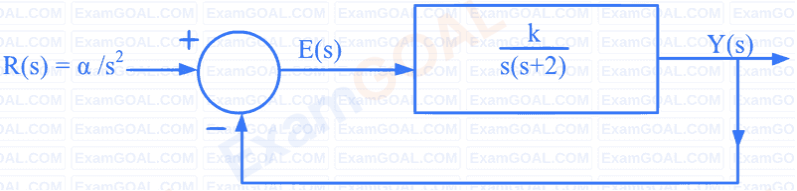
The impulse response of the closed-loop system decays faster than e^{-t} if _____.

(A) $1 \leq K \leq 5$ (B) $7 \leq K \leq 21$ (C) $-4 \leq K \leq -1$ (D) $-24 \leq K \leq -6$

Q24. GATE ECE 2023

Open solution

A closed loop system is shown in the figure where $k > 0$ and $\alpha > 0$. The steady state error due to a ramp input $(R(s) = \alpha/s^2)$ is given by



(A) $\frac{2\alpha}{k}$ (B) $\frac{\alpha}{k}$ (C) $\frac{\alpha}{2k}$ (D) $\frac{\alpha}{4k}$

Q25. GATE ECE 2022

Open solution

Two linear time-invariant systems with transfer functions

$G_1(s) = \frac{10}{s^2+s+1}$ and $G_2(s) = \frac{10}{s^2+s\sqrt{10}+10}$

have unit step responses $y_1(t)$ and $y_2(t)$, respectively. Which of the following statements is/are true?

(A) $y_1(t)$ and $y_2(t)$ have the same percentage peak overshoot.

(B) $y_1(t)$ and $y_2(t)$ have the same steady-state value.

(C) $y_1(t)$ and $y_2(t)$ have the same damped frequency of oscillation.

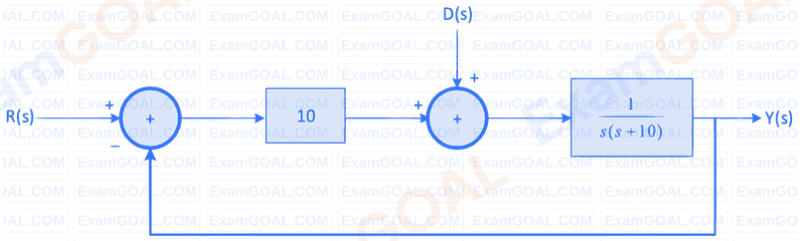
(D) $y_1(t)$ and $y_2(t)$ have the same 2% settling time.

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Q26. GATE ECE 2022

Open solution

The block diagram of a closed-loop control system is shown in the figure. R(s), Y(s), and D(s) are the Laplace transforms of the time-domain signals r(t), y(t), and d(t), respectively. Let the error signal be defined as e(t) = r(t) − y(t). Assuming the reference input r(t) = 0 for all t, the steady-state error e(∞), due to a unit step disturbance d(t), is _____ (rounded off to two decimal places).



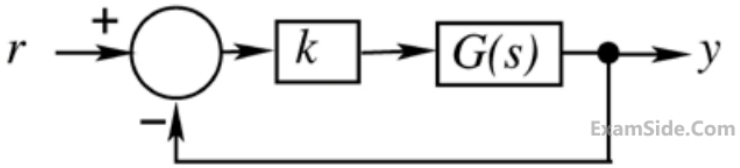
Q27. GATE ECE 2016 Set 2

Open solution

In the feedback system shown below

$$G(S) = \frac{1}{(s^2 + 2s)}$$

The step response of the closed-loop system should have minimum settling time and have no overshoot.



The required value of gain k to achieve this is _____.

Q28. GATE ECE 2016 Set 1

Open solution

The open-loop transfer function of a unity-feedback control system is

$$G(S) = \frac{K}{s(s + 2)}$$

For the peak overshoot of the closed-loop system to a unit step input to be 10%, the value of K is _____.

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Q29. GATE ECE 2015 Set 2

Open solution

The output of a standard second–order system for a unit step input is given as

$$y(t) = 1 - \frac{2}{\sqrt{3}} e^{-t} \cos\left(\sqrt{3}t - \frac{\pi}{6}\right)$$

The transfer function of the system is

(A) $\frac{2}{(s+2)\left(s+\sqrt{3}\right)}$ (B) $\frac{1}{s^2+2s+1}$

(C) $\frac{3}{s^2+2s+3}$ (D) $\frac{4}{s^2+2s+4}$

Q30. GATE ECE 2015 Set 1

Open solution

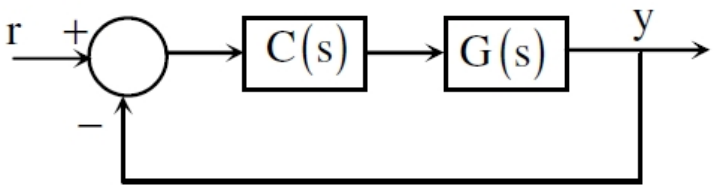
The damping ratio of a series RLC circuit can be expressed as

(A) $\frac{R^2C}{2L}$ (B) $\frac{2L}{R^2C}$ (C) $\frac{R}{2}\sqrt{\frac{C}{L}}$ (D) $\frac{2}{R}\sqrt{\frac{L}{C}}$

Q31. GATE ECE 2014 Set 1

Open solution

For the following feedback system $G(s) = \frac{1}{(s+1)(s+2)}$.The 2% settling time of the step response is required to be less than 2 seconds.



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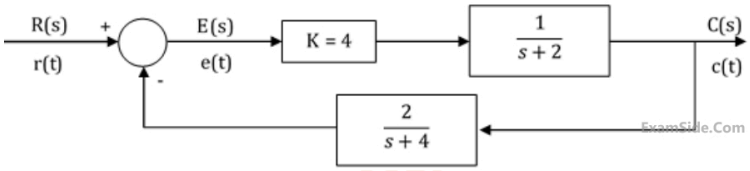
Which one of the following compensators C(s) achieves this?

(A) $3\left(\frac{1}{s+5}\right)$ (B) $5\left(\frac{0.03}{s}+1\right)$ (C) 2(s+4) (D) $4\left(\frac{s+8}{s+3}\right)$

Q32. GATE ECE 2014 Set 3

Open solution

The steady state error of the system shown in the figure for a unit step input is _____.

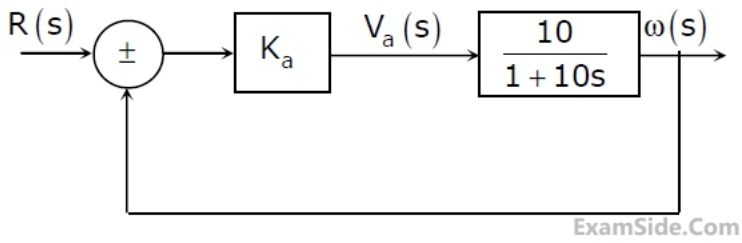


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Q33. GATE ECE 2013

The open-loop transfer function of a dc motor is given as $\frac{\omega(s)}{V_a(s)} = \frac{10}{1+10s}$, when connected in feedback as shown below, the approximate value of K_a that will reduce the time constant of the closed-loop system by one hundred times as compared to that of the open-loop system is



Open solution

(A) 1 (B) 5 (C) 10 (D) 100

Q34. GATE ECE 2009

The unit step response of an under-damped second order system has steady state value of -2. Which one of the following transfer function has these properties?

(A) $\frac{-2.24}{s^2 + 2.59s + 1.12}$ (B) $\frac{-3.82}{s^2 + 1.91s + 1.91}$

(C) $\frac{-2.24}{s^2 - 2.59s + 1.12}$ (D) $\frac{-3.82}{s^2 - 1.91s + 1.91}$

Open solution

Q35. GATE ECE 2008

A linear, time-invariant, causal continuous time system has a rational transfer function with simple poles at s=-2 and s=-4, and one simple zero at s=-1. A unit step u(t) is applied at the input of the system. At steady state, the output has constant value of 1. The impulse response of this system is

(A) [exp(-2t) + exp(-4t)]u(t) (B) [-4exp(-2t) + 12exp(-4t) - exp(-t)]u(t)

(C) [-4exp(-2t) + 12exp(-4t)]u(t) (D) [-0.5exp(-2t) + 1.5exp(-4t)]u(t)

Open solution

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Q36. GATE ECE 2008

Open solution

Group I lists a set of four transfer functions. **Group II** gives a list of possible step responses y(t). Match the step responses with the corresponding transfer functions

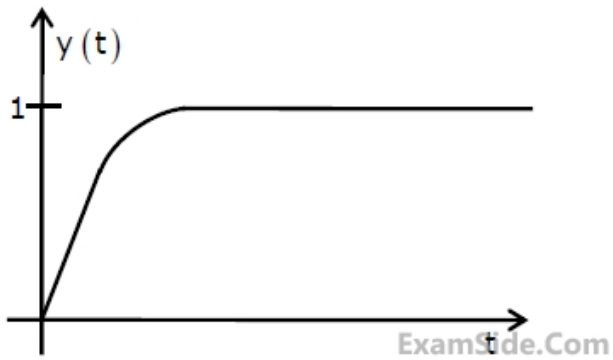
Group I

$$P = \frac{25}{s^2 + 25} \qquad Q = \frac{36}{s^2 + 20s + 36}$$

$$R = \frac{36}{s^2 + 12s + 36} \qquad S = \frac{49}{s^2 + 7s + 49}$$

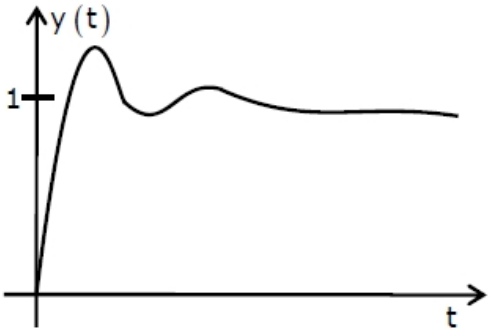
Group II

(1)



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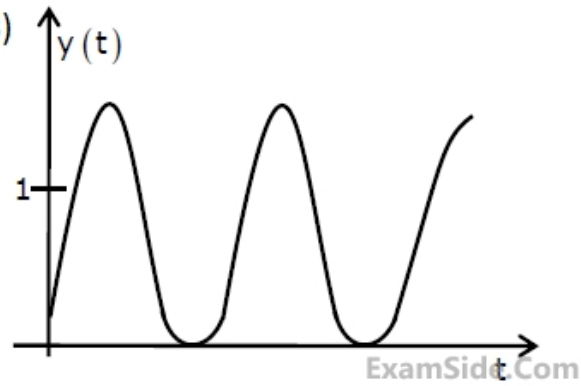
(2)



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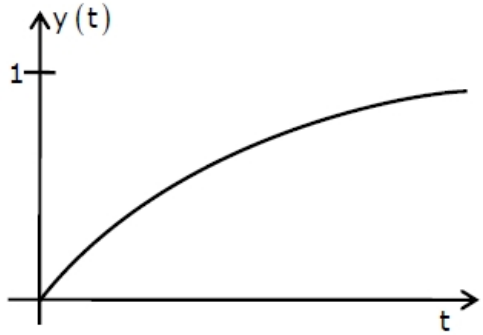
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(3)



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(4)



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(A) $\begin{matrix} P & Q & R & S \\ 3 & 1 & 4 & 2 \end{matrix}$

(B) $\begin{matrix} P & Q & R & S \\ 3 & 2 & 4 & 1 \end{matrix}$

(C) $\begin{matrix} P & Q & R & S \\ 2 & 1 & 4 & 3 \end{matrix}$

(D) $\begin{matrix} P & Q & R & S \\ 3 & 4 & 1 & 2 \end{matrix}$

Q37. GATE ECE 2007

The frequency response of a linear, time-invariant system is given by $H(f) = \frac{5}{1 + j10\pi f}$ .The step response of the system is:

Open solution

(A) $5\left(1 - e^{-5t}\right)u(t)$

(B) $5\left(1 - e^{-\frac{t}{5}}\right)u(t)$

(C) $\frac{1}{5}\left(1 - e^{-5t}\right)u(t)$

(D) $\frac{1}{5}\left(1 - e^{-\frac{t}{5}}\right)u(t)$

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Q38. GATE ECE 2007

The transfer function of a plant is

$$T(s) = \frac{5}{(s+5)(s^2+s+1)}$$

The second-order approximation of T (s) using dominant pole concept is:

(A) $\frac{1}{(s+5)(s+1)}$ (B) $\frac{5}{(s+5)(s+1)}$

(C) $\frac{5}{(s^2+s+1)}$ (D) $\frac{1}{(s^2+s+1)}$

Open solution

Q39. GATE ECE 2006

The unit impulse response of a system is:

$$h(t) = e^{-t},\ t \geq 0$$

For this system, the steady-state value of the output for unit step input is equal to

(A) -1 (B) 0 (C) 1 (D) ∞

Open solution

Q40. GATE ECE 2006

The unit-step response of a system starting from rest is given by

$$c(t) = 1 - e^{-2t} \text{ for } t \geq 0$$

The transfer function of the system is:

(A) $\frac{1}{1+2s}$ (B) $\frac{2}{2+s}$ (C) $\frac{1}{2+s}$ (D) $\frac{2s}{1+2s}$

Open solution

Q41. GATE ECE 2005

A ramp input applied to an unity feedback system results in 5% steady state error. The type number and zero frequency gain of the system are respectively.

(A) 1 and 20 (B) 0 and 20 (C) 0 and 1/20 (D) 1 and 1/20

Open solution

Q42. GATE ECE 2005

In the derivation of expression for peak percent overshoot,

$$M_p = exp\left(\frac{-\pi\xi}{\sqrt{1-\xi^2}}\right) \times 100\%$$

.Which one of the following conditions is NOT required?

(A) System is linear and time invariant.

(B) The system transfer function has a pair of complex conjugate poles and no zeroes.

(C) There is no transportation delay in the system.

(D) The system has zero initial conditions.

Open solution

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Q43. GATE ECE 2004

A causal system having the transfer function $G(s) = \frac{1}{s+2}$ is excited with 10u(t). The time at which the output reaches 99% of its steady state value is

(A) 2.7 sec (B) 2.5 sec (C) 2.3 sec (D) 2.1 sec

Open solution

Q44. GATE ECE 2004

A system described by the following differential equation

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = x(t)$$

is initially at rest. For input x(t) = 2u(t), the output y(t) is

(A) $\left(1 - 2e^{-t} + e^{-2t}\right)u(t)$ (B) $\left(1 + 2e^{-t} - 2e^{-2t}\right)u(t)$

(C) $\left(0.5 + e^{-t} + 1.5e^{-2t}\right)u(t)$ (D) $\left(0.5 + 2e^{-t} + 2e^{-2t}\right)u(t)$

Open solution

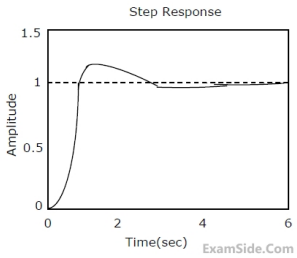
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Q45. GATE ECE 2003

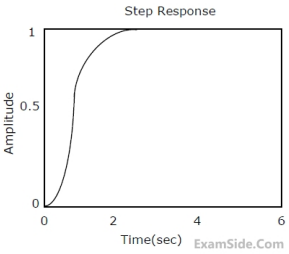
A second-order system has the transfer function $\frac{C(s)}{R(s)} = \frac{4}{s^2+4s+4}$. With r(t) as the unit-step function, the response c(t) of the system is represented by

Open solution

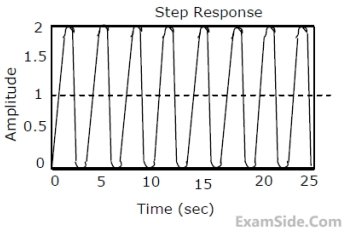
(A)



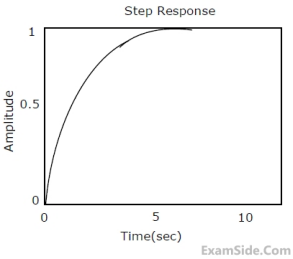
(B)



(C)



(D)



Q46. GATE ECE 2002

The transfer function of a system is $G(s) = \frac{100}{(s+1)(s+100)}$. For a unit step input to the system the approximate settling time for 2% criterion is

(A) 100 sec (B) 4 sec (C) 1 sec (D) 0.01 sec

Open solution

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Q47. GATE ECE 1999

If the closed-loop transfer function T(s) of a unity negative feedback system is given by

$$T(s) = \frac{a_{n-1}s + a_n}{s^n + a_1 s^{n-1} + \ldots\ldots + a_{n-1}s + a_n}$$

then the steady state error for a unit ramp input is

(A) $\frac{a_n}{a_{n-1}}$ (B) $\frac{a_n}{a_{n-2}}$ (C) $\frac{a_{n-2}}{a_{n-2}}$ (D) zero

Open solution

Q48. GATE ECE 1998

The unit impulse response of a linear time invariant system is the unit step function u(t). For t>0, the response of the system to an excitation e^{-at} u(t), a > 0 will be

(A) ae^{-at} (B) $(1/a)(1 - e^{-at})$ (C) $a(1 - e^{-at})$ (D) $1 - e^{-at}$

Open solution

Q49. GATE ECE 1994

The response of an LCR circuit to a step input is
(a) Over damped
(b) Critically damped
(c) Oscillatory
If the transfer function has

(1) poles on the negative real axis
(2) poles on the imaginary axis
(3) multiple poles on the positive real axis
(4) poles on the positive real axis
(5) multiple poles on the negative real axis

Open solution

Q50. GATE ECE 1994

Match the following with List-1 with List-2

List-1

(a) Very low response at very high frequencies.
(b) Overshoot
(c) Synchro-control transformer output

List-2

(1) Low pass systems
(2) Velocity damping
(3) Natural frequency
(4) Phase-sensitive modulation
(5) Damping ration

Open solution

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Q51. GATE ECE 1991

A second order system has a transfer function given by $G(s) = \frac{25}{s^2 + 8s + 25}$ .If the system, initially at rest is subjected to a unit step input at t = 0, the second peak in response will occur at

(A) π sec (B) $\frac{\pi}{3}$ sec (C) $\frac{2\pi}{3}$ sec (D) $\frac{2\pi}{3}$ sec

Open solution

Q52. GATE ECE 1991

A unity feedback control system has the open loop transfer function $G(s) = \frac{4(1+2s)}{s^2(s+2)}$ .If the input to the system is a unit ramp, the steady-state error will be

(A) 0 (B) 0.5 (C) 2 (D) Infinity.

Open solution

Q53. GATE ECE 1990

The steady state error of a stable 'type 0' unity feedback system for a unit step function is

(A) 0 (B) $\frac{1}{1+K_p}$ (C) ∞ (D) $\frac{1}{K_p}$

Open solution

Q54. GATE ECE 1988

A critically damped, continuous-time, second order system, when sampled, will have (in Z domain)

(A) A simple pole (B) double pole on real axis
(C) Double pole on imaginary axis (D) a pair of complex conjugate poles

Open solution

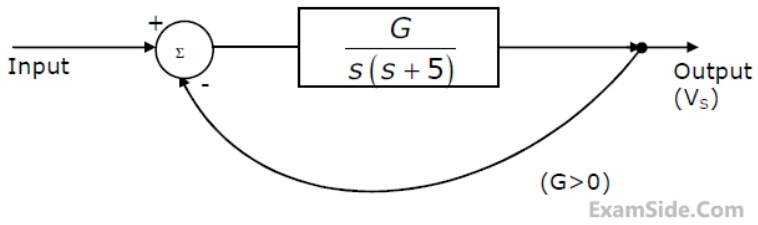
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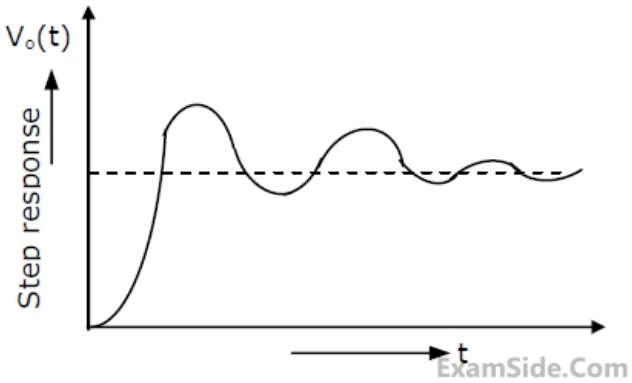
Marks 5 2 questions

Q55. GATE ECE 2000

Open solution

The block diagram of a feedback system is shown in Figure.





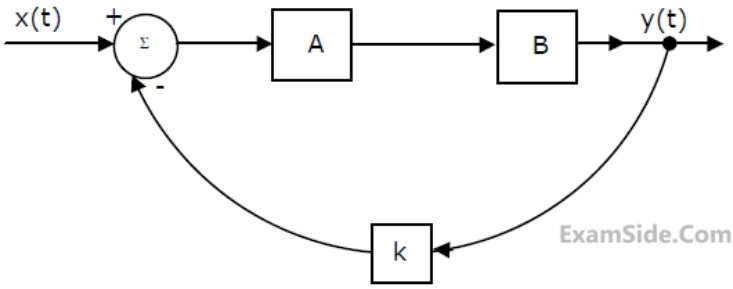
(a) Find the closed loop transfer function.
(b) Find the minimum value of G for which the step response of the system would exhibit an overshoot, as shown in Figure.
(c) For G equal to twice this minimum value, find the time period T indicated in Figure.

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Q56. GATE ECE 1997

Open solution

Following fig. shows the block diagram representation of control system. The system in block A has an impulse response h(t) = e^{-t} u(t). The system in block B has an impulse response h(t) = e^{-2t} u(t). The block 'K' amplifies its inputs by a factor k. For the overall system with input x(t) and output y(t).



(a) Find the transfer function $\frac{y(s)}{x(s)}$ when k =1.
(b) Find the impulse response when k = 0.
(c) Find the values of k for which the system becomes unstable.

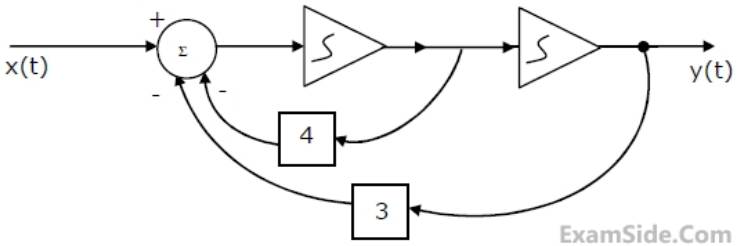
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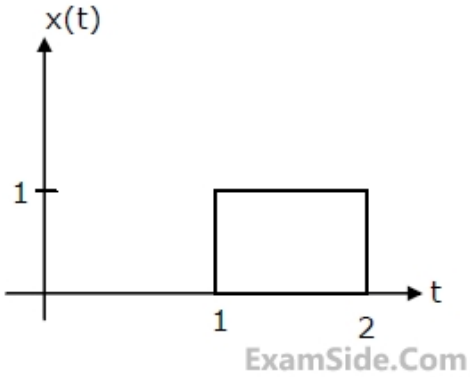
Marks 8 2 questions

Q57. GATE ECE 2000

Open solution

For the linear, time-invariant system whose block diagram is shown in Fig. with input x(t) and output y(t),





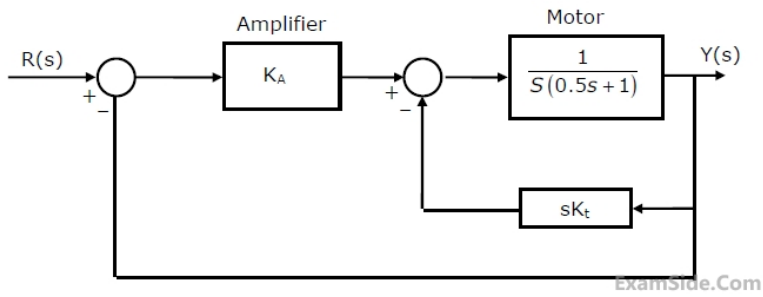
(a) Find the transfer function.
(b) For the step response of the system [i.e. find y(t) when x(t) is a unit step function and the initial conditions are zero]
(c) Find y(t), if x(t) is as shown in Fig. and the initial conditions are zero.

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Q58. GATE ECE 1992

Open solution

Block diagram model of a position control system is shown in figure.



(a) In absence of derivative feedback (K_t=0) , determine damping ratio of the system for amplifier gain K_A=5. Also find the steady state error to unit ramp input.

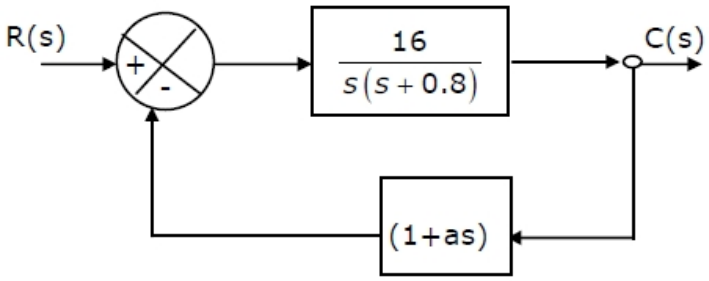
(b) Find suitable values of the parameters K_A and K_t so that damping ratio of the system is increased to 0.7 without affecting the steady state error as obtained in part (a).

Marks 10 1 questions

Q59. GATE ECE 1998

Open solution

Consider the system shown in Fig. Determine the value of 'a' such that the damping ratio is 0.5. Also obtain the values of the rise time 't_r' and maximum overshoot 'M_p' in its step response.



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Answer Key

Marks 1

1. **D**	2. **1**	3. **1**	4. **A**	5. **4**
6. **C**	7. **38.15 to 38.16**	8. **0**	9. **A**	10. **C**
11. **D**	12. **C**	13. **C**	14. **B**	15. **A**
16. **A**	17. **B**	18. **C**	19. **A**	20. **B**
21. **A**	22. $\pm j\omega_n$			

Marks 2

23. **A**	24. **A**	25. **A**	26. **-0.1**	27. **1**
28. **1.2 to 1.3**	29. **D**	30. **C**	31. **C**	32. **0.5**
33. **C**	34. **B**	35. **C**	36. **D**	37. **B**
38. **D**	39. **C**	40. **B**	41. **A**	42. **C**
43. **C**	44. **A**	45. **B**	46. **B**	47. **D**
48. **B**	49. **(a) - (1) , (b) - (5) , (c) - (2)**	50. **(a) - (1) , (b) - (5) , (c) - (4)**	51. **A**	52. **A**
53. **B**	54. **B**			

Marks 5

55. **(a) Closed loop transfer function=** $\frac{G}{s^2+3s+G}$ **(b) G = 6.25 (c) T= 1.96 sec**	56. **(a)** $\frac{y(s)}{x(s)}=\frac{1}{s^2+3s+3}$ **(b) Impulse response h(t)=** $\left(e^{-t}-e^{-2t}\right)u(t)$ **(c) When k<-2 system is unstable.**

Marks 8

57. **(a)** $\frac{Y(s)}{X(s)}=\frac{1}{s^2+4s+3}$ **(b)** $y(t)=\frac{1}{3}-\frac{1}{2}e^{-t}+\frac{1}{6}e^{-3t}$ **(c)** $y(t)=L^{-1}[Y(s)]=[1]-[2]$ $Where$ $[1]=\frac{1}{3}u(t-1)-\frac{1}{2}e^{-(t-1)}u(t-1)+\frac{1}{6}e^{-3(t-1)}u(t-1)$ $[2]=\frac{1}{3}u(t-2)-\frac{1}{2}e^{-(t-2)}u(t-2)+\frac{1}{6}e^{-3(t-2)}u(t-2)$	58. **(a)** $\frac{C(s)}{R(s)}=\frac{10}{s^2+2s+10}$ **Steady state error for unit Ramp=0.2 (b)** K_A **=25 and** K_t **= 4**

Marks 10

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59.

a = 0.2
t_r = 0.6053 sec
M_p = 0.163

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